The Role of Spatial Structure in Human and Model-Based Categorization

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Introduction + Background

- Embodied cognition theories propose spatial frameworks guide abstract thought and decision-making [1,2].
- Humans naturally use familiar spatial structures such as rows, grids, and axes to process both spatial and nonspatial information [3].

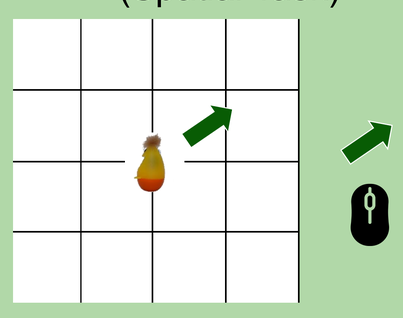
Open Question: Can spatial memory representations **facilitate** categorization even when categories lack inherent spatial meaning?

Hypothesis: Simpler spatial layouts (1D) should be easier to leverage for categorization than more complex, interdigitated layouts, resulting in higher categorization accuracy and faster response time.

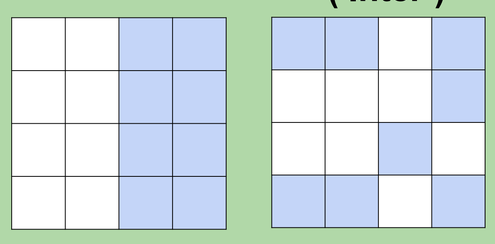
Methods

Paradigm:

Object-location association (Spatial Task)

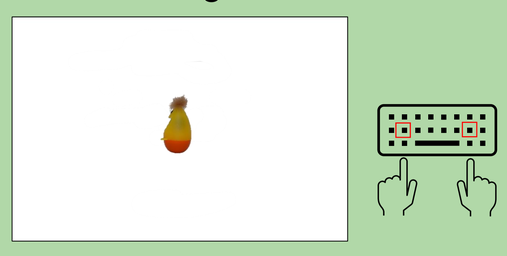


Spatial Layouts:

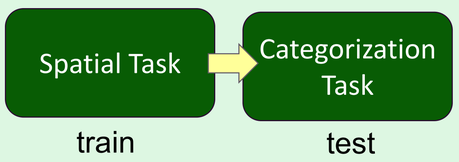


Categories: A B

Categorization Task



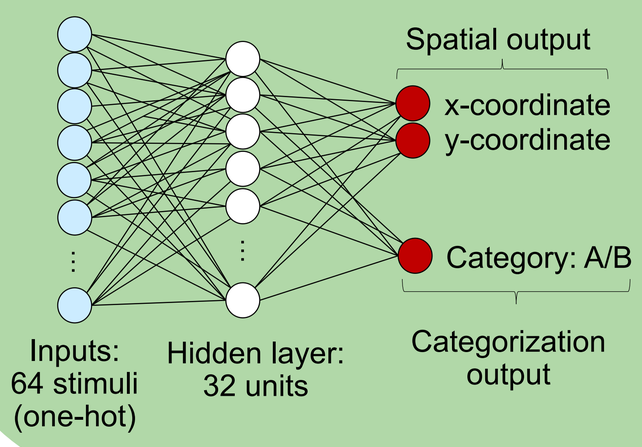
This sequence is repeated per phase (4 phases total):



Metrics: accuracy, response time (RT)

Modeling:

Linear Dual-Head Multi-layer Perceptron (MLP)



- Loss function: MSE for spatial, BCE for categorization
- Varied weight initializations: rich vs lazy regimes
- SDG optimizer
- Online learning
- Modeled ground truth data of all participants in pilot sample N=33

Discussion

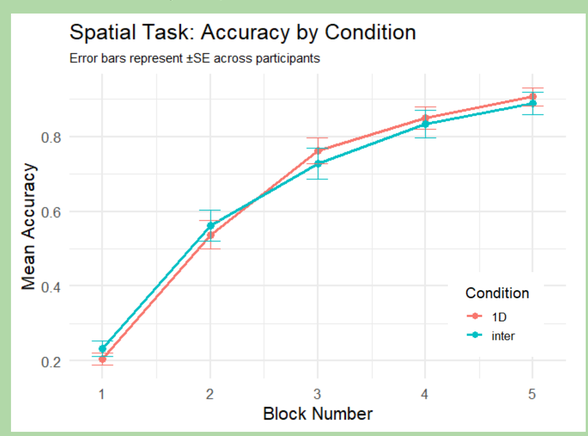
- We observe a main effect of the simpler, more separable spatial condition (1D) with higher categorization accuracy than the more complex (inter) condition. The congruency effect further shows that participants exploit alignment between memory and motor responses.
- MLPs learned more structured and separable embeddings in the 1D than the interdigitated condition

Next steps:

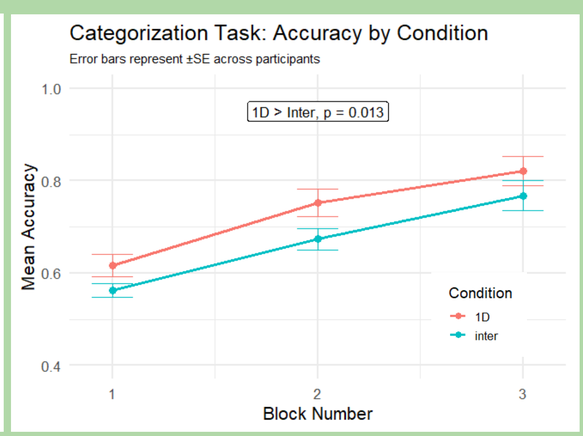
- Collect larger sample and/or modify task design for fMRI scanner to confirm main effects
- Explore alternative models

Behavioral Results

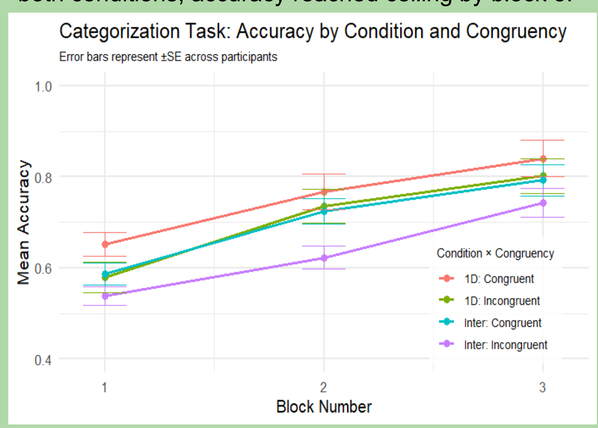
Behavioral analyses performed on N=29



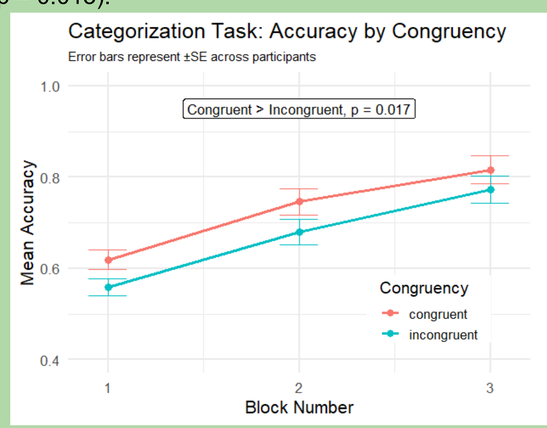
Participants learned spatial locations over blocks in both conditions; accuracy reached ceiling by block 5.



Categorization was more accurate in 1D than Inter (p = 0.013).

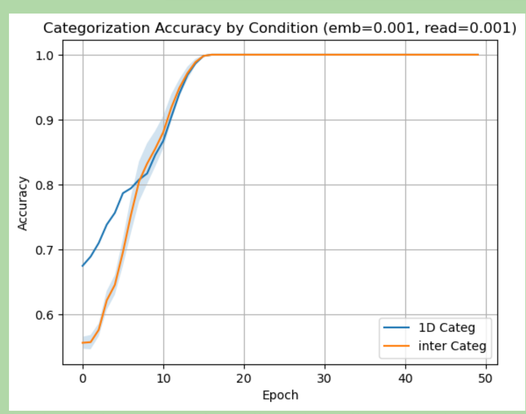


Both 1D and Inter showed congruency trends, but no significant interaction of condition x congruency

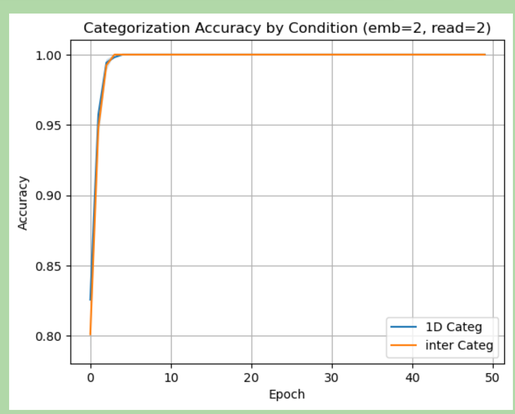


Categorization was more accurate in Congruent trials than Incongruent trials (p = 0.017).

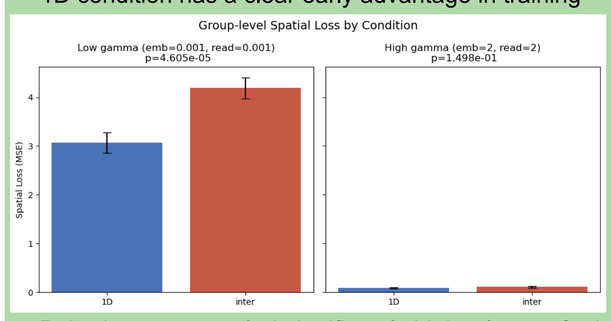
Modeling Results

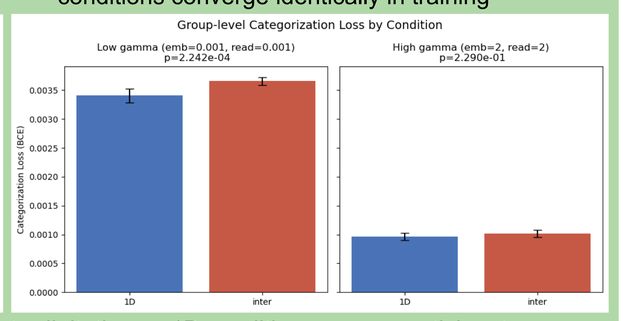


In the rich regime (small embedding weights), the 1D condition has a clear early advantage in training



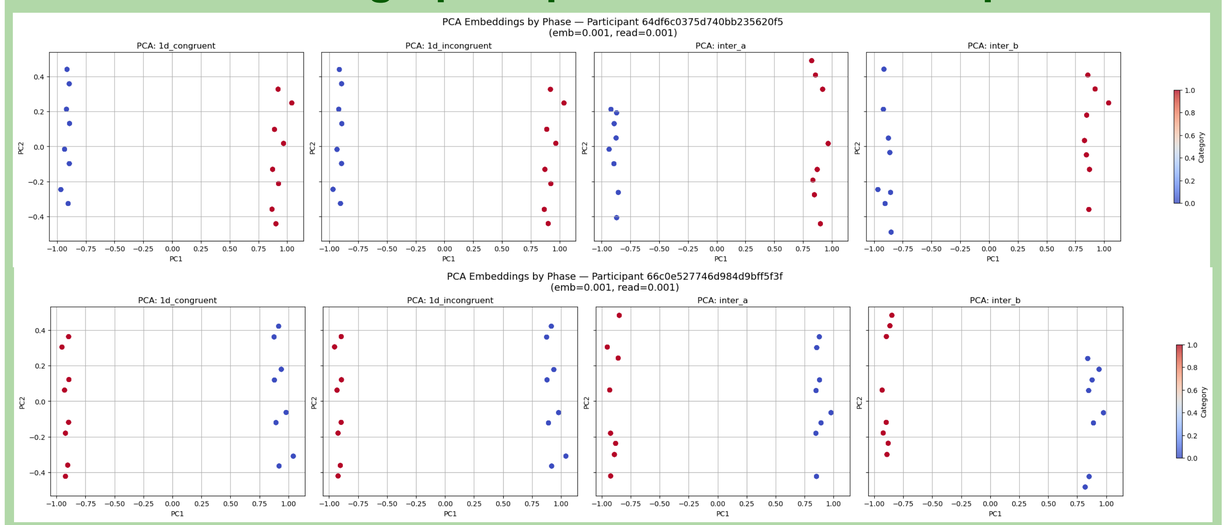
In the lazy regime (large embedding weights), both conditions converge identically in training



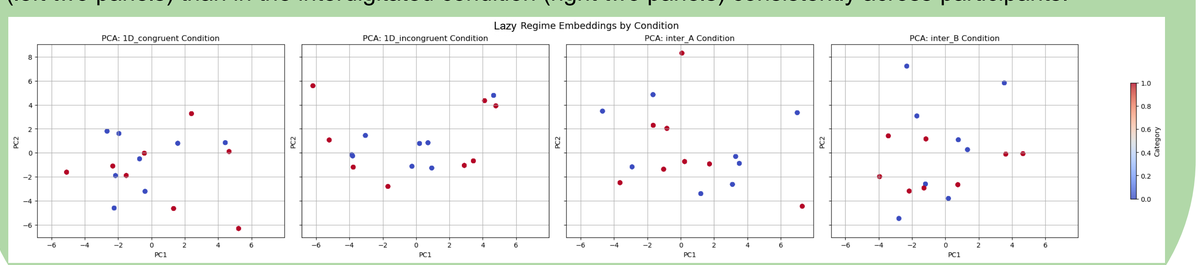


Paired t-tests revealed significantly higher losses for interdigitation vs 1D conditions across participants, indicating that the model consistently learned spatial mappings more efficiently in the 1D condition.

Latent Embedding Space per Task Phase examples:



The latent shared embedding space retains much of the initial spatial grid structure better in the 1D condition (left two panels) than in the interdigitated condition (right two panels) consistently across participants.



The latent shared embedding space in the lazy regime does not retain the initial spatial grid structure in any condition consistently across participants.

References

[1] Gärdenfors, P. (2000). Conceptual spaces: The geometry of thought. MIT Press.
[2] Gattis, M. (Ed.). (2001). Spatial schemas and abstract thought. MIT Press.
[3] Shah, P. (Ed.). (2005). The Cambridge handbook of visuospatial thinking (1. publ). Cambridge University Press.

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